

SOFTWARE ENGINEERING

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #58 – Media, Events & Community

Digital Art Commission & Watermarking Portal

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1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall apply a dynamic diagonal overlay watermark to every WIP draft uploaded by a Digital Artist before the draft is displayed to the Client Buyer.	High	Pass: For each of 20 supported test images, the client portal displays only a watermarked preview containing the commission ID and buyer identifier, while the original remains inaccessible. Fail: Any preview lacks the watermark or an original draft is retrievable by the client.	Protects unfinished artwork from unauthorized use and preserves the supplied FR-001 intent.
FR-002	Functional	The system shall allow a Client Buyer to submit a creative brief containing scope, reference images, dimensions, file-format needs, deadline, budget, and milestone terms.	High	Pass: With all mandatory valid fields, one commission request with a unique ID is stored and shown within 3 seconds. Fail: A request is created with a missing mandatory field, an unsupported reference file, or without a commission ID.	Creates an unambiguous, traceable specification against which the artist can quote and deliver.
FR-003	Functional	The system shall allow the assigned Digital Artist to accept a commission, record agreed milestones, and upload versioned WIP drafts against an open milestone.	High	Pass: An assigned artist can accept a seeded request and upload two drafts that appear as distinct, timestamped versions under the selected milestone. Fail: An unassigned artist can accept or upload, a version is overwritten, or the upload is not linked to the milestone.	Supports controlled collaboration and preserves a reviewable history of work in progress.

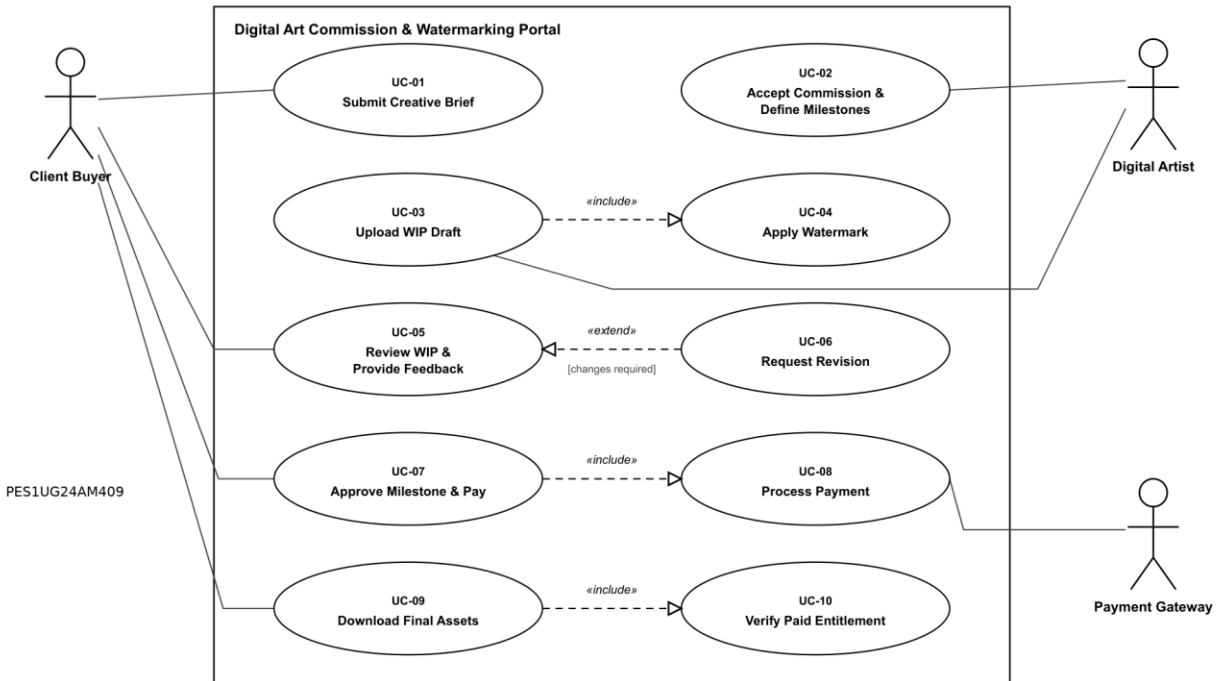
1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-004	Functional	The system shall allow the Client Buyer to review a watermarked WIP draft, add contextual feedback, request a revision, or approve the associated milestone.	High	Pass: A buyer reviewing a seeded draft can submit feedback or approve the milestone, and the selected action, comment, user ID, and timestamp are visible to the artist within 5 seconds. Fail: A non-owner can review the draft, an action lacks an audit record, or status remains stale.	Provides a formal approval loop and reduces disputes about requested changes and milestone acceptance.
FR-005	Functional	The system shall unlock the final high-resolution source files for the Client Buyer only after the required milestone payment is verified as successful.	High	Pass: Before payment, every final-file request returns 403 without file data; after a successful seeded payment, exactly one entitlement is recorded and the buyer can download the correct final asset. Fail: Access is granted before verification, the wrong file is served, or duplicate entitlements are created.	Enforces the platform's payment-for-delivery exchange and protects the artist's high-value source files.
NFR-001	Nonfunctional Performance & Security	The system shall deliver final assets only through signed download URLs that expire 60 minutes after issuance and shall generate an authorized URL within 2 seconds.	High	Pass: In a 10,000-request test with 300 concurrent users, at least 95% of authorized URLs are generated within 2 seconds; all URLs expire at 60 minutes $\pm$ 5 seconds; and altered, expired, or unauthorized URLs return 401/403 without file data. Fail: Any threshold is missed.	Makes the supplied signed-URL requirement measurable while limiting unauthorized redistribution.
NFR-002	Nonfunctional Availability & Auditability	The system shall provide at least 99.9% monthly availability, excluding announced maintenance, and retain immutable audit events for brief, draft, approval, payment, and download actions for 365 days.	Medium	Pass: Monthly monitoring reports availability of at least 99.9%, and an audit test confirms that 100% of 1,000 seeded actions record actor, action, commission ID, and UTC timestamp and remain queryable after a simulated 365-day retention cycle. Fail: Either condition is not met.	Maintains dependable collaboration and evidence for resolving authorship, approval, payment, and access disputes.

2. UML Use-Case Diagram

UML Use-Case Diagram

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3. Use-Case Flow Specification

Use Case ID / Name: UC-03 - Upload WIP Draft

Actors: Digital Artist (primary), Client Buyer (supporting)

Trigger: The artist is ready to submit a WIP draft for an open commission milestone.

Preconditions: The commission is active; the artist is assigned; the target milestone is open; the artist is authenticated; the file uses a supported image format.

Postconditions (Success): The original is stored privately, a versioned watermarked preview is linked to the milestone, and the buyer is notified. Minimum guarantee: no unwatermarked draft is exposed if processing fails.

Main Success Scenario

1. Digital Artist opens the active commission and selects the open milestone.
2. System displays the milestone brief, accepted formats, maximum file size, and prior WIP versions.
3. Digital Artist selects 'Upload WIP Draft', chooses an image, and enters an optional version note.
4. System validates artist assignment, milestone state, file type, file size, and malware-scan result.
5. System stores the original draft in private asset storage and creates a new immutable version record.
6. System applies a diagonal watermark containing the commission ID and buyer identifier to a derived preview.
7. System verifies that the watermark is present and that no public or buyer-facing link targets the original file.
8. System publishes the watermarked preview to the commission workspace and timestamps the submission.
9. System notifies the Client Buyer that a WIP draft is available for review and confirms successful submission to the Digital Artist.

Alternate Flow – A1: Validation or Watermarking Failure at Step 4 or 7

- System rejects the draft or marks the new version as 'Processing Failed' and records the reason.
- System removes all buyer-facing access to the failed preview and retains the original only in private storage according to policy.
- System informs the Digital Artist of the specific correction required and returns to Step 3; the Client Buyer is not notified and no unwatermarked asset is exposed.